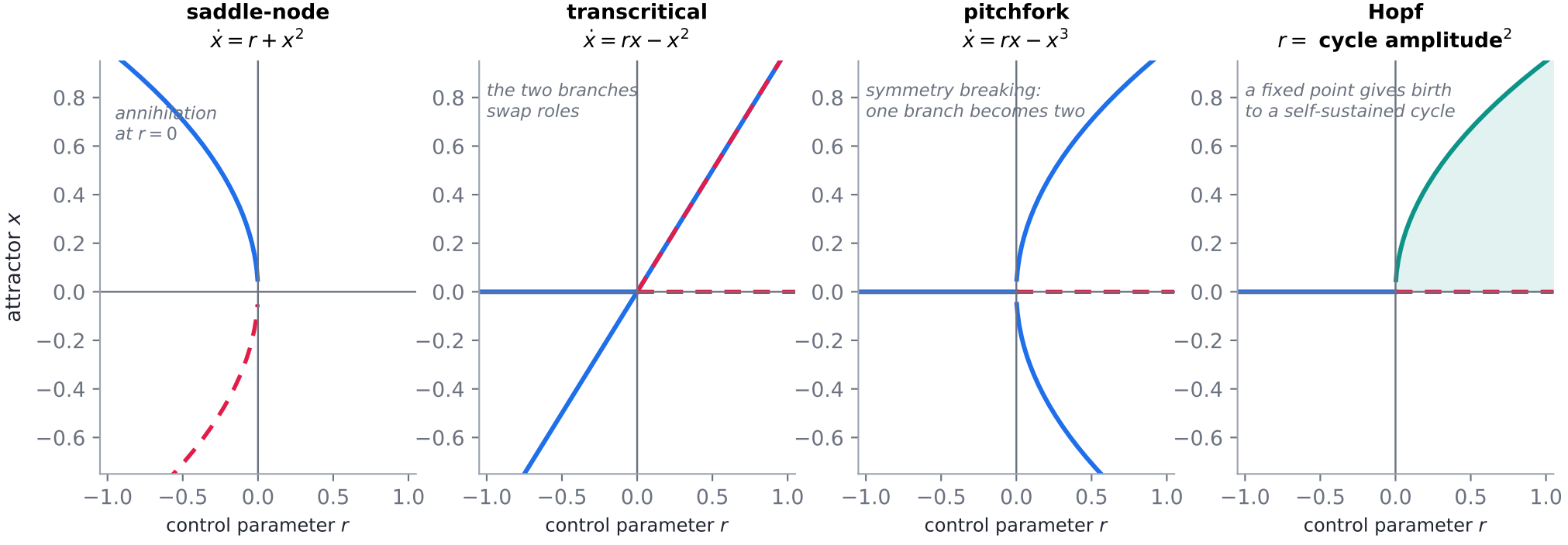
Bifurcations: stable branch solid, unstable branch dashed
saddle-node
$\dot{x} = r + x^2$
annihilation
at $r = 0$
transcritical
$\dot{x} = rx - x^2$
the two branches
swap roles
pitchfork
$\dot{x} = rx - x^3$
symmetry breaking:
one branch becomes two
Hopf
$r =$ cycle amplitude^2
a fixed point gives birth
to a self-sustained cycle
attractor x
control parameter r
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−1.0
−0.5
0.0
0.5
1.0